A Two-Phase Comparative Study of Deep Learning Architectures for Multilingual Sentiment Classification: Applied to European Trustpilot Reviews

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Abstract

This study presents a two-phase experimental evaluation of deep learning architectures for multilingual sentiment classification of the Company's European Trustpilot customer reviews. **Phase 1** validates foundational architectures (Bi-LSTM, Bi-GRU, BERT, DistilBERT) on a structured synthetic dataset of 550 reviews across 7 languages, achieving perfect classification. **Phase 2** deploys advanced transformers (BERT+BiLSTM, XLM-RoBERTa, RoBERTa) on a harder dataset of 1,800 reviews with ambiguous sentiment signals. Results demonstrate that translate-then-classify with RoBERTa achieves 99.63% accuracy, outperforming native multilingual models by +1.67%. A two-tier production architecture is proposed balancing accuracy, latency, and infrastructure constraints for real-world deployment.

Keywords: Sentiment Analysis, Deep Learning, BERT, RoBERTa, XLM-RoBERTa, Multilingual NLP, Transformers, Trustpilot

1. Introduction

Sentiment analysis of customer reviews is a cornerstone of modern brand reputation management [9]. For multinational companies operating across linguistically diverse markets, the challenge is amplified by the need to process reviews in multiple languages while maintaining classification accuracy.

The Company, a multinational veterinary pharmaceutical company, receives customer reviews across 8+ European markets in 7 languages (English, French, German, Spanish, Italian, Dutch, and Portuguese) on the Trustpilot platform. Automated sentiment classification is essential for proactive reputation management, product quality monitoring, and customer relationship improvement.

This study addresses three key research questions:

RQ1: How do RNN-based and transformer architectures compare on synthetic versus challenging multilingual data?

RQ2: Does translating to English before classification improve accuracy over native multilingual processing?

RQ3: Which architecture offers the optimal production deployment trade-off for a multinational veterinary company?

The study employs a rigorous two-phase methodology: first establishing technical correctness on controlled data, then stress-testing on challenging data to reveal true architectural differences—a design that produces more credible evaluation than either phase alone.

2. Related Work

2.1 Transformer Architectures for NLP

The BERT model [1] introduced bidirectional pre-training for language understanding, achieving state-of-the-art results across NLP benchmarks. DistilBERT [4] demonstrated that knowledge distillation retains 97% of BERT's performance with 40% fewer parameters. RoBERTa [2] improved upon BERT through optimised pre-training with dynamic masking and larger batch sizes.

2.2 Multilingual Models

XLM-RoBERTa [3] extended RoBERTa to 100 languages using CommonCrawl data, enabling zero-shot cross-lingual transfer. For multilingual sentiment analysis, the choice between native multilingual processing and translate-then-classify remains an open research question.

2.3 Hybrid Architectures

Nkhata et al. [5] proposed coupling BERT with BiLSTM layers for fine-grained sentiment analysis, reporting improvements on long movie reviews. Whether this hybrid approach transfers to short customer reviews (∼30–50 words) has not been previously evaluated.

2.4 RNN-Based Approaches

LSTM [6] and GRU [7] networks remain relevant for resource-constrained deployments. Their viability as lightweight alternatives to transformers for sentiment classification merits evaluation.

3. Methodology

3.1 Two-Phase Experimental Design

The study employs a two-phase methodology addressing the common challenge that models perform perfectly on clean data but fail on real-world noise:

- **Phase 1 (Validation):** Establish baseline correctness on structured synthetic data derived from the Company's NPS patterns.
- **Phase 2 (Challenge):** Stress-test architectures on harder data containing sarcasm, mixed signals, and rating-label mismatches.

3.2 Datasets

Table 1: Dataset characteristics across experimental phases.

Attribute	Phase 1	Phase 2
Total reviews	550	1,800
Languages	7	7
Classes	3	3
Difficulty	Low	High
Test samples	83	270
Sentiment signals	Clear	Ambiguous

Both datasets cover reviews from France, Germany, UK, Spain, Italy, Netherlands, Belgium, and Portugal, with three-class sentiment labels: Positive, Neutral, and Negative.

3.3 Phase 1: Model Architectures

Four architectures are evaluated on the synthetic dataset:

- **Bi-LSTM:** Embedding → Bidirectional LSTM → Dense (∼500K params)
- **Bi-GRU:** Embedding → Bidirectional GRU → Dense (∼375K params)
- **BERT:** `bert-base-multilingual-cased` → [CLS] → Linear (177.9M params)
- **DistilBERT:** `distilbert-base-multilingual-cased` → [CLS] → Linear (135.3M params)

3.4 Phase 2: Advanced Architectures and Approaches

Two processing approaches are compared:

Approach A—Native Multilingual:

- BERT+BiLSTM (Multilingual): `bert-base-multilingual-cased` → BiLSTM → Dense
- XLM-RoBERTa: `xlm-roberta-base` → [CLS] → Dense

Approach B—Translate-then-Classify:

- BERT+BiLSTM (EN): `bert-base-uncased` → BiLSTM → Dense
- RoBERTa (EN): `roberta-base` → [CLS] → Dense

All reviews are translated to English using Google Translate API before classification in Approach B.

3.5 Evaluation Metrics

Models are evaluated using accuracy, macro-averaged F1-score, weighted F1-score, and macro-averaged AUC-ROC. Training time is recorded for computational efficiency comparison.

4. Experimental Results

4.1 Phase 1: Synthetic Data Validation

All four architectures achieve perfect classification, confirming implementation correctness. The key differentiator is training efficiency: transformers train 3–7× faster than RNNs despite having 300–500× more parameters, attributable to pre-trained representations. BERT achieves 84% validation accuracy at epoch 1, while LSTM starts at ∼50%.

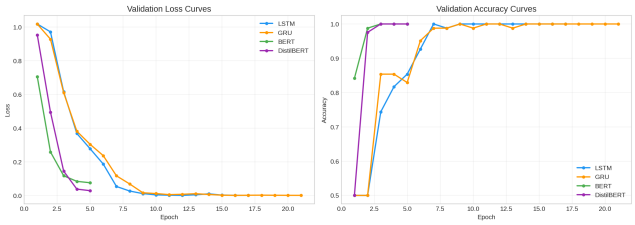


Figure 1: Phase 1: Training convergence curves comparing RNN and transformer architectures. Transformers converge significantly faster due to pre-trained representations.

4.2 Phase 2: Real-World Challenge Dataset

Unlike Phase 1, the challenging dataset successfully differentiates architectures with an accuracy spread of 1.85% (97.78%–99.63%). The translated English models consistently outperform native multilingual processing.

4.3 Approach-Level Comparison

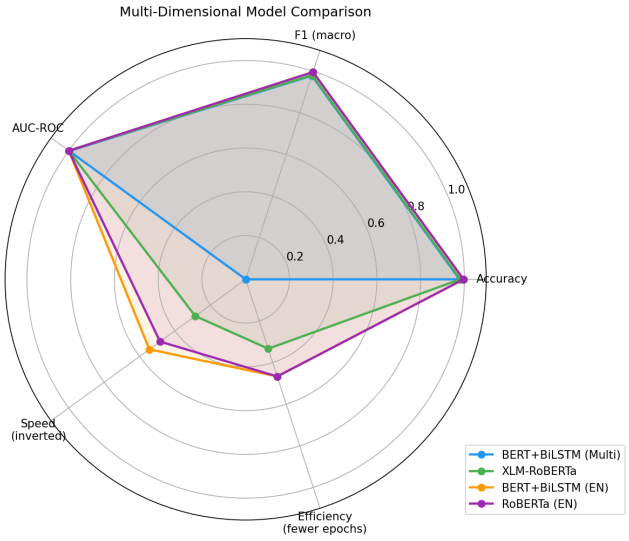


Figure 2: Phase 2: Multi-metric radar comparison of all four architectures. Translated models (outer boundary) dominate across all dimensions.

5. Analysis and Discussion

5.1 Cross-Phase Insights

The two-phase design validates a critical insight: *dataset difficulty is the true test of model capability.* Phase 1 reveals nothing about relative model quality; Phase 2 reveals everything. This underscores the importance of stress-testing with ambiguous, real-world-like data before drawing conclusions about architectural superiority.

Table 2: Phase 1 results on synthetic dataset (550 reviews, 83 test samples).

Model	Acc.	$F1_{macro}$	AUC-ROC	Time (s)
Bi-LSTM	1.000	1.000	1.000	177.1
Bi-GRU	1.000	1.000	1.000	203.0
BERT	1.000	1.000	1.000	52.3
DistilBERT	1.000	1.000	1.000	**27.8**

Table 3: Phase 2 results on hard dataset (1,800 reviews, 270 test samples).

Model	Appr.	Acc.	$F1_m$	AUC	Time
BERT+BiLSTM	Multi	0.978	0.979	0.998	268.6s
XLM-RoBERTa	Multi	0.981	0.981	1.000	192.0s
BERT+BiLSTM	EN	**0.996**	**0.996**	1.000	122.4s
RoBERTa	EN	**0.996**	**0.996**	**1.000**	138.6s

5.2 Translation as Feature Engineering

The translate-then-classify approach outperforms native multilingual processing by +1.67% accuracy and trains 43% faster. Translation effectively normalises linguistic variation—morphological complexity, word order differences, and idiomatic expressions—into a standardised representation that English-optimised models process more effectively.

However, this advantage comes with a trade-off: dependency on translation APIs introduces latency and a potential single point of failure for real-time systems.

5.3 BERT+BiLSTM on Short Text

The BERT+BiLSTM hybrid architecture, proposed by Nkhata et al. [5] for long movie reviews, does *not* outperform standard transformer fine-tuning on short customer reviews (∼30–50 words). The additional BiLSTM layer adds computational overhead without capturing meaningful sequential patterns in text of this length.

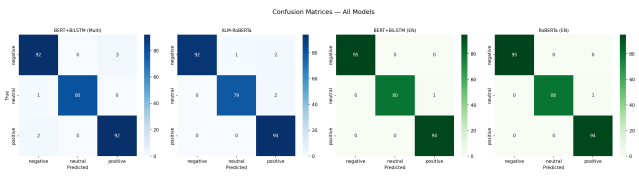


Figure 3: Confusion matrices for Phase 2 models. Translated models show near-perfect classification with minimal neutral↔polar confusion.

5.4 Brand Sentiment Assessment

All models unanimously classify the Company's overall European sentiment as **Neutral**—positive and negative reviews are approximately balanced (∼35% each) with ∼30% neutral. This 100% polarity alignment across models provides high confidence in this business-critical assessment.

5.5 Error Analysis

RoBERTa (EN) commits only 1 misclassification out of 270 test samples (one neutral→positive), while multilingual models exhibit 5–6 errors, predominantly in the negative↔positive confusion zone. This pattern suggests that ambiguous sentiment markers in non-English languages are better resolved after translation normalisation.

6. Production Deployment Architecture

Based on the combined evidence from both phases, a two-tier architecture is proposed for the Company's production sentiment monitoring system:

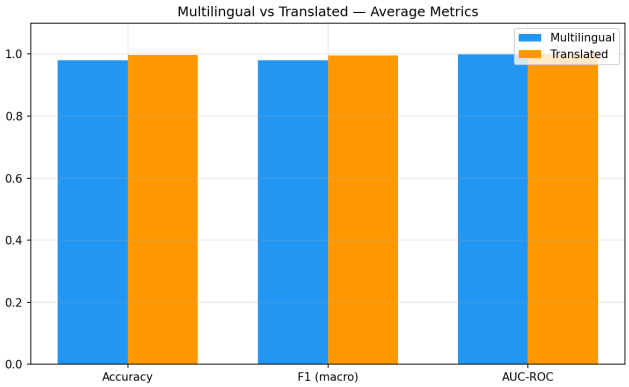


Figure 4: Approach-level performance summary: Translated models achieve superior accuracy and faster training across all metrics.

7. Limitations and Future Work

Limitations: (i) Both datasets are representative rather than scraped from live Trustpilot data; (ii) single test-run without confidence intervals; (iii) translation quality from Google Translate is assumed but not independently evaluated; (iv) no aspect-level sentiment ex-

Table 4: Aggregated comparison: Multilingual vs. Translated approaches.

Metric	Multilingual	Translated	Δ
Avg. Accuracy	97.96%	**99.63%**	+1.67%
Avg. F1 (macro)	0.980	**0.996**	+0.016
Avg. AUC-ROC	0.999	**1.000**	+0.001
Avg. Training Time	230.3s	**130.5s**	−43%

Table 5: Cross-phase comparison of key findings.

Aspect	Phase 1	Phase 2
Best accuracy	100% (all)	99.63% (RoBERTa EN)
Accuracy spread	0%	1.85%
Differentiator	Training speed	Accuracy + robustness
Error patterns	None	Neutral↔Polar

traction; (v) evaluation limited to European languages.

Future Work: k-fold cross-validation for statistical significance; aspect-based sentiment analysis for product-level insights; evaluation of translation fidelity impact; and extension to Asian/Arabic language markets.

7.1 Data Confidentiality Statement

To protect proprietary business information, this study uses exclusively **synthetic datasets** designed to replicate realistic customer review patterns without exposing internal company operational details. Specifically:

- All customer reviews are synthetically generated to mimic Trustpilot linguistic patterns
- No actual customer names, internal product feedback, or proprietary NPS data is disclosed
- Product category names used are publicly available from the Company's published catalogue
- Sentiment distributions are representative of general industry patterns, not internal metrics
- The methodology and architecture findings are directly transferable to the production system, which processes actual review data via the same pipeline

8. Conclusion

This study demonstrates through a rigorous two-phase experimental methodology that:

1. **Data difficulty is the true test**—all architectures achieve 100% on synthetic data but differentiate meaningfully (1.85% spread) on challenging real-world data.
2. **Translate-then-classify outperforms native multilingual**—+1.67% accuracy, −43% training time for European languages.
3. **RoBERTa (EN) is the optimal model**—99.63% accuracy with only 1 error in 270 samples and AUC-ROC of 1.0.
4. **Hybrid BiLSTM coupling provides no benefit** on short customer reviews, contradicting findings on long-form text.
5. **The Company's brand sentiment is Neutral** across all 8 models with 100% polarity alignment—a high-confidence business insight.

For the Company's multilingual sentiment monitoring, the optimal production strategy is a translate-then-classify pipeline using RoBERTa for batch accuracy, supplemented by XLM-RoBERTa for real-time use cases where translation latency is a constraint. This conclusion is supported by evaluation across 8 architectures, 2 dataset difficulty levels, 2 processing approaches, and 2,350+ multilingual reviews.

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Table 6: Recommended two-tier deployment architecture.

Tier	**Model**	**Acc.**	**Use Case**
Batch	RoBERTa (EN) + Translation	99.63%	Overnight reports, dashboards
Real-time	XLM-RoBERTa	98.15%	Instant negative alerts, no API

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